

Erdős Problem #684

EP-684

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

For $0 \leq k \leq n$ write $\binom{n}{k} = uv$ where the only primes dividing u are in $[2, k]$ and the only primes dividing v are in $(k, n]$.

Let $f(n)$ be the smallest k such that $u > n^2$. Give bounds for $f(n)$.

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>